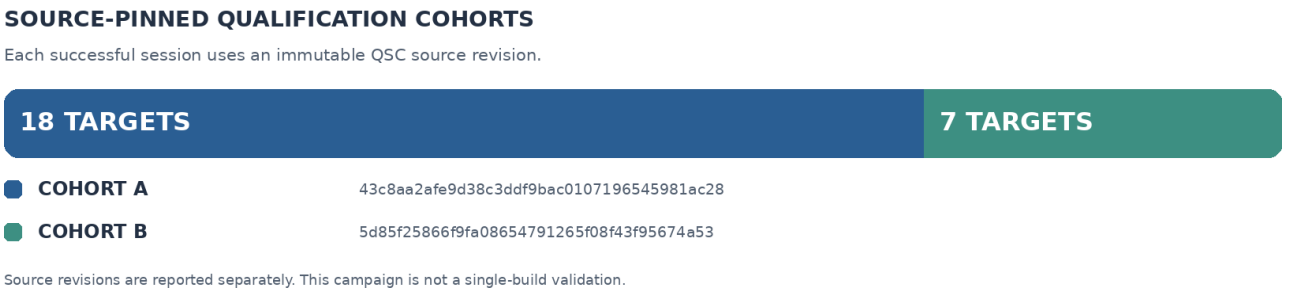
QRCS

QSC Cryptographic Library NIST ACVTS Demo Qualification Evidence

NOVEMBER 2026 | TECHNICAL ASSURANCE REPORT



25 independent targets. 86 submitted vector sets.

The completed qualification campaign returned 25 server-reported passing Demo sample verdicts using two explicitly identified source-pinned QSC revisions.

Document identifier: QRCS-QSC-ACVTS-QR-2026-11-001

Prepared and issued by QRCS Corporation, Ottawa, Canada

Evidence status: Verified from archived GitHub Actions client artifacts

Classification: Public technical qualification evidence

This report concerns NIST ACVTS Demo sample tests. It does not assert CAVP or CMVP certification.

Executive summary

QRCS Corporation completed a source-pinned testing campaign against the NIST Automated Cryptographic Validation Testing System demonstration environment. The consolidated evidence register contains 25 separate successful manual sample test sessions and 86 submitted algorithm vector sets. In each session, the client summary records PASS_DEMO_SAMPLE and server_passed=true.

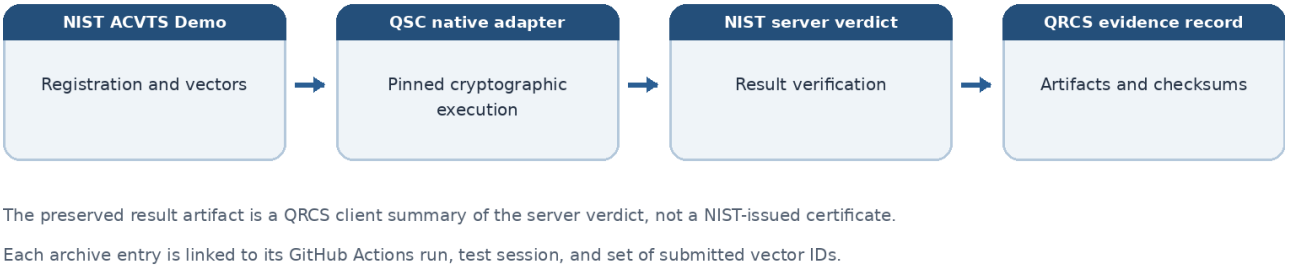


Figure 1. Evidence collection and verification chain.

Key results

Measure	Verified outcome
Planned algorithm qualification targets	25 / 25 successful Demo sessions
Submitted vector sets	86 across the 25 sessions
Evidence artifacts preserved	25 original Actions ZIP files and 25 normalized reports
QSC source cohorts	18 earlier targets / 7 later targets
Final ECC qualification	KAS-ECC-SSC, session 788041, PASS_DEMO_SAMPLE
Formal certificate issuance	Not asserted or issued by these sample sessions

Scope boundary. An ACVTS Demo sample is technical conformance evidence, not a public CAVP algorithm certificate or a FIPS 140-3 module validation. NIST production validation is an additional laboratory-led process.

Evidence interpretation

The results are tied to the tested source revisions, registered capabilities, available native implementations, selected Linux scalar compilation paths and NIST sample vector sets. The campaign does not establish exhaustive coverage of every instruction-set path, operating system, QSC API or future source revision.

Campaign scope and methodology

1. Native cryptographic execution

The independently maintained QSC-ACVP qualification repository checked out QSC read-only at immutable commit references. Native QSC objects were built on GitHub-hosted Linux runners and exercised through thin protocol adapters. Responses were submitted to NIST ACVTS Demo by

explicitly authorized workflows. The retained client result is a sanitized JSON record, not a full cryptographic transcript.

2. NIST session procedure

- Register the selected algorithms and capabilities with isSample=true.
- Retrieve server-generated vector sets, respecting retry and readiness instructions.
- Execute the native QSC implementation and prepare responses using the designated adapters.
- Submit responses to the session and obtain the server-reported verdict.
- Archive the sanitized verdict, workflow reference, tested source pin and cryptographic digest of the evidence files.

3. Offline qualification controls

The staged program also exercised preregistration checks, malformed-vector handling, public-key validation, independent cryptographic oracles, DRBG prerequisites, credential-safe reporting and GitHub-hosted offline regression suites. These offline checks supplement, but are not counted as, the 25 successful NIST Demo sessions.

Source-revision traceability

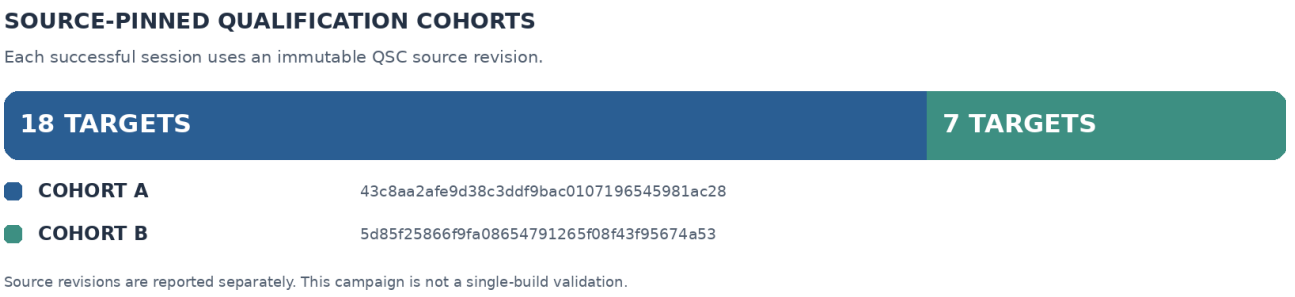


Figure 2. Two immutable source revisions, not a single source-wide certification claim.

Cohort	Qualified targets	QSC commit reference
A | Earlier campaign	18	43c8aa2afe9d38c3ddf9bac0107196545981ac28
B | HCG and ECC extension	7	5d85f25866f9fa08654791265f08f43f95674a53

The source pins were recovered from the workflow definitions at the corresponding GitHub Actions head revisions. Selected original job logs independently confirm checkout of both commits. Each record in the evidence manifest identifies its cohort and the associated Actions run.

Important. This campaign demonstrates 25 passing targets over two tested QSC revisions. It should not be described as 25 tests against one current executable or as a validation of all QSC build variants.

Controlled execution and integrity

The workflows used protected ACVTS Demo credentials, a repository-controlled native adapter, automated source-pin enforcement and manually approved server contact. The public evidence archive excludes JWT access tokens, private certificate keys, TOTP seed material and raw

cryptographic session secrets. Original artifact ZIPs have been copied byte-for-byte into the evidence bundle and are accompanied by SHA-256 digests.

Qualified algorithm target inventory

Table 1 lists each of the 25 independently completed manual qualification targets. Session and vector-set identifiers are taken directly from the preserved result artifacts. A run URL resolves to the original GitHub Actions job while the retention period remains active.

No.	Target	Session	Sets	Source
01	SHA2-256	787624	1	A
02	SHA2-384	787625	1	A
03	SHA2-512	787626	1	A
04	SHA3-256	787627	1	A
05	SHA3-512	787637	1	A
06	SHAKE-128	787639	1	A
07	SHAKE-256	787641	1	A
08	HMAC-SHA2-256	787644	2	A
09	HMAC-SHA2-384	787646	2	A
10	HMAC-SHA2-512	787648	2	A
11	AES-ECB	787654	1	A
12	AES-CBC	787656	1	A
13	AES-CTR	787664	1	A
14	AES-GCM	787665	2	A
15	PBKDF	787672	4	A
16	ML-KEM	787673	6	A
17	ML-DSA	787674	5	A
18	SLH-DSA	787676	4	A
19	HMAC_DRBG-SHA2-512	787980	3	B
20	KDA-HKDF	787994	9	B
21	ECDSA	787996	8	B
22	DetECDSA	787997	6	B
23	EDDSA	788004	8	B
24	XECDH	788040	6	B
25	KAS-ECC-SSC	788041	9	B

Table 1. The listed targets include repeat testing of prerequisite algorithms inside later sessions. Vector-set totals count distinct submissions across all 25 run artifacts.

Server verdict reconciliation

All 25 original result objects report PASS_DEMO_SAMPLE and server_passed=true. Twenty-four reports also contain the final vector_verdict_statuses value ["passed"]. One early SHA2-256 sample has a distinct, explicitly retained inconsistency: the overall verdict is passing, while the captured vector_verdict_statuses value is ["incomplete"].

Early SHA2-256 record

Evidence field	Recorded value
NIST session	787624
Submitted vector set	4148700

Evidence field	Recorded value
Overall status	PASS_DEMO_SAMPLE
server_passed	true
vector_verdict_statuses	["incomplete"]
Follow-up evidence	SHA2-256 passed in sessions 787644 and 787994

The original anomalous value remains unchanged in the published archive. Subsequent independent same-session prerequisite runs contain ["passed"] vector verdicts for SHA2-256. Consequently the completed SHA2-256 target is not presented as supported solely by the anomalous status field.

Evidence completeness and residual limitations

- The result files are sanitized QSC client summaries containing the server-reported session verdict. They do not contain NIST-signed certificate artifacts.
- The evidence manifest contains the Actions run IDs, session identifiers, all vector-set IDs and their algorithm and mode labels, original archived ZIP hashes and normalized JSON hashes.
- GitHub Actions source runs can expire. The separately downloaded ZIP bundle preserves the 25 retained client artifacts, subject to further independent backup by QRCS.
- The sample suites, operating environment and source cohort should be stated when quoting the result. The report does not imply automatic applicability to later QSC releases or hardware acceleration paths.
- For formal certification, accredited laboratory testing and production ACVTS validation remain distinct stages.

Qualification evidence by functional category

Category	Targets	Count
Hash and extensible-output functions	SHA2-256/384/512, SHA3-256/512, SHAKE-128/256	7
Message authentication	HMAC-SHA2-256/384/512	3
AES block-cipher modes	ECB, CBC, CTR, GCM	4
Password-based derivation	PBKDF	1
Post-quantum public key algorithms	ML-KEM, ML-DSA, SLH-DSA	3
Deterministic random bit generation	HMAC_DRBG-SHA2-512	1
Key derivation	KDA-HKDF	1
Elliptic curve and Edwards algorithms	ECDSA, DetECDSA, EDDSA, XECDH, KAS-ECC-SSC	5

Table 2. Classification of the 25 qualification targets.

Technical significance

The campaign verifies that the tested implementations produced server-accepted results across classical, post-quantum and elliptic-curve algorithm families, with a deterministic HMAC_DRBG prerequisite used in the later elliptic-curve sessions. It also provides reproducible provenance from GitHub-hosted workflows to the associated QSC source commits.

For customers, the important result is traceability. An auditor can identify the implementation version, algorithm operation, NIST Demo test session, submitted vector set and retained server-reported verdict. This is stronger than an undocumented assertion that algorithms have passed local test vectors.

Detailed evidence index and vector sets

Table 3 lists the vector-set identifiers carried in each saved artifact. Algorithm and operation labels are recorded in full in the accompanying JSON manifest. Long lists are presented without truncating test-set coverage.

01 SHA2-256 | Session 787624

Vector-set IDs: 4148700

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43321340135

02 SHA2-384 | Session 787625

Vector-set IDs: 4148701

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43328100214

03 SHA2-512 | Session 787626

Vector-set IDs: 4148702

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43328944584

04 SHA3-256 | Session 787627

Vector-set IDs: 4148703

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43329645114

05 SHA3-512 | Session 787637

Vector-set IDs: 4148741

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43338490656

06 SHAKE-128 | Session 787639

Vector-set IDs: 4148744

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43339494189

07 SHAKE-256 | Session 787641

Vector-set IDs: 4148747

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43340356998

08 HMAC-SHA2-256 | Session 787644

Vector-set IDs: 4148782, 4148783

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43341213710

09 HMAC-SHA2-384 | Session 787646

Vector-set IDs: 4148786, 4148787

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43341894638

10 HMAC-SHA2-512 | Session 787648

Vector-set IDs: 4148790, 4148791

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43342406504

11 AES-ECB | Session 787654

Vector-set IDs: 4148849

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43350776946

12 AES-CBC | Session 787656

Vector-set IDs: 4148851

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43351680223

13 AES-CTR | Session 787664

Vector-set IDs: 4148911

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43357212151

14 AES-GCM | Session 787665

Vector-set IDs: 4148912, 4148913

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43359628131

15 PBKDF | Session 787672

Vector-set IDs: 4148920, 4148921, 4148922, 4148923

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43361501531

16 ML-KEM | Session 787673

Vector-set IDs: 4148924, 4148925, 4148926, 4148927, 4148928, 4148929

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43363695106

17 ML-DSA | Session 787674

Vector-set IDs: 4148930, 4148931, 4148932, 4148933, 4148934

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43365698700

18 SLH-DSA | Session 787676

Vector-set IDs: 4148936, 4148937, 4148938, 4148939

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43367485294

19 HMAC_DRBG-SHA2-512 | Session 787980

Vector-set IDs: 4151645, 4151646, 4151647

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43501732560

20 KDA-HKDF | Session 787994

Vector-set IDs: 4151871, 4151872, 4151873, 4151874, 4151875, 4151876, 4151877, 4151878, 4151879

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43518311133

21 ECDSA | Session 787996

Vector-set IDs: 4151888, 4151889, 4151890, 4151891, 4151892, 4151893, 4151894, 4151895

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43526926643

22 DetECDSA | Session 787997

Vector-set IDs: 4151896, 4151897, 4151898, 4151899, 4151900, 4151901

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43534002049

23 EDDSA | Session 788004

Vector-set IDs: 4151950, 4151951, 4151952, 4151953, 4151954, 4151955, 4151956, 4151957

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43537644321

24 XECDH | Session 788040

Vector-set IDs: 4152263, 4152264, 4152265, 4152266, 4152267, 4152268

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43545312767

25 KAS-ECC-SSC | Session 788041

Vector-set IDs: 4152269, 4152270, 4152271, 4152272, 4152273, 4152274, 4152275, 4152276, 4152277

Evidence: https://github.com/QRCS-CORP/QSC-ACVP/actions/runs/43547438305

Evidence integrity and archival controls

Public evidence archive structure

- evidence-manifest.json: detailed, structured target/session/vector-set index, source cohorts, sanitized report paths, run URLs and SHA-256 digests.
- evidence-index.csv: concise target-level register suitable for independent import.
- sanitized_server_results/: 25 normalized JSON summaries of the server responses as reported by the qualification client.
- archived_actions_artifacts/: byte-for-byte copies of the 25 original GitHub Actions artifact ZIPs.
- SHA256SUMS.txt: cryptographic digests of every archived evidence file.
- README.md: scope, usage and known limitations.

Recommended verification procedure

- Download the evidence archive through the QRCS published evidence record.
- Compute SHA-256 hashes and compare against SHA256SUMS.txt.
- Inspect evidence-manifest.json for the run, session, vector and QSC source revision of interest.
- Compare the normalized JSON with the untouched original ZIP, and review the original GitHub Actions record if still retained.
- Apply the qualification scope boundary when issuing derivative statements or integrating this report into a customer dossier.

Public communication controls

Appropriate wording	Avoid
Passed 25 NIST ACVTS Demo sample qualification targets	NIST-certified QSC library
Source-pinned algorithm-conformance evidence	FIPS 140-3 validated cryptographic module
Server-reported PASS_DEMO_SAMPLE verdicts	NIST issued 25 certificates
Documented qualification for identified source versions	All QSC operating systems and CPU paths certified

The recommended GitHub badge should identify the Demo sample context explicitly and link to the full report. It is a QRCS-authored evidence-status mark, not a NIST certification insignia.

Formal validation pathway and supporting sources

NIST CAVP production ACVTS access is restricted to accredited laboratories. To seek an actual certificate, QRCS should select an accredited CST or 17ACVT laboratory and agree on the exact algorithms, source modules, implementation boundary, operating environments and production test scope. The Demo administrator can clarify whether additional non-sample Demo validation and metadata registration would be useful preparatory work.

Public source references

NIST Automated Cryptographic Validation Protocol: https://pages.nist.gov/ACVP/

NIST ACVTS Demo and Prod access requirements: https://csrc.nist.gov/projects/cryptographic-algorithm-validation-program/how-to-access-acvts

NIST ACVP JSON protocol, testSessions and isSample: https://pages.nist.gov/ACVP/draft-fussell-acvp-spec.html

GitHub Actions artifact and run retention policy: https://docs.github.com/en/repositories/managing-your-repositorys-settings-and-features/enabling-features-for-your-repository/managing-github-actions-settings-for-a-repository

QSC-ACVP public qualification repository: https://github.com/QRCS-CORP/QSC-ACVP

QSC library source repository: https://github.com/QRCS-CORP/QSC

Document control and closing record

Field	Value
Document number	QRCS-QSC-ACVTS-QR-2026-11-001
Publisher	QRCS Corporation
Date of issue	November 13, 2026 (America/Toronto)
Classification	Public technical qualification evidence
Evidence collection	25 original NIST Demo result artifacts
Total submitted vector sets	86
Tested QSC revisions	43c8aa2a... and 5d85f258... (full hashes in Table 2)
Status	Evidence compiled and verified against saved sanitized artifacts
Certificate status	No CAVP or CMVP certificate asserted

Summary of accomplished work

The QSC qualification campaign completed 25 planned NIST ACVTS Demo sample targets across 86 submitted vector sets and two immutable source revisions. The record spans SHA-2/SHA-3/SHAKE, HMAC, AES, PBKDF, ML-KEM, ML-DSA, SLH-DSA, HMAC_DRBG, KDA-HKDF, ECDSA, deterministic ECDSA, EdDSA, XECDH and KAS-ECC-SSC.

The archived results, machine-readable evidence manifest, run history and documented limitations provide a credible technical basis for customer evaluation and independent review. The evidence should continue to be maintained as new QSC source versions, compilation profiles and algorithm parameters are qualified.

Follow-on action. Maintain a versioned evidence register, link the report from the main QSC README, archive all sanitized result ZIPs in a durable location, and consult an accredited cryptographic testing laboratory before undertaking formal CAVP validation.